Structure-Based Druggability Assessment of Two PDAC Targets: ERBB4 and CEL

BIOT 6900 — Module 4 Assignment
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Disease: Pancreatic ductal adenocarcinoma (PDAC)

Summary of structural assessment

Gene	UniProt	Mod-2 rank	mean pLDDT	% < 50	Domains	PDB	RMSD (Å)	Best pocket	Verdict	Druggability call
ERBB4	Q15303	4	73.1	26.9	2	2R4B	2.82	0.894	keep	Structure supports a small-molecule pocket
CEL	P19835	11	82.6	16.2	1	1F6W	1.35	0.925	keep	Possible but uncertain

Inter-domain PAE: ERBB4 28.7 Å (two domains, uncertain arrangement); CEL n/a (single confident domain, residues 21–552).

1. Which parts of each target can I trust?

ERBB4

The ERBB4 AlphaFold model has a mean pLDDT of 73.1, which suggests moderate overall confidence, but this average does not tell the whole story. About 26.9% of the model has a pLDDT below 50, while 43.8% is above 90. Much of the lower confidence comes from regions such as the long C-terminal tail, which is intrinsically disordered. The kinase domain is much better modelled than the whole-protein average suggests, making it more useful for assessing potential drug-binding sites.

The PAE within pocket 93 is 5.4 Å, and its mean pLDDT is 80.9, indicating that the pocket itself is reasonably well supported. However, the inter-domain PAE is 28.7 Å, meaning the relative positions of the extracellular and intracellular regions are uncertain. ERBB4 is a membrane receptor, and AlphaFold does not model it within its natural membrane environment, which can affect the predicted orientation of the transmembrane helix and neighbouring domains.

The low-confidence regions also have biological explanations. Residues 1–24 form the signal peptide, which is removed during protein maturation. Residues 637–652 and 688–708 lie around the membrane-to-intracellular transition and juxtamembrane region, where flexibility and the absence of a membrane environment can contribute to uncertainty. Residues 1025–1308 belong to the intrinsically disordered C-terminal tail, which acts as a docking platform for signalling proteins. Since this tail does not maintain one fixed structure, its low pLDDT can reflect its normal biological behaviour rather than a prediction failure. Overall, I would place more confidence in the kinase domain and its binding pocket than in the orientation of the full receptor.

FIGURE: ERBB4 and CEL pLDDT profile along the sequence

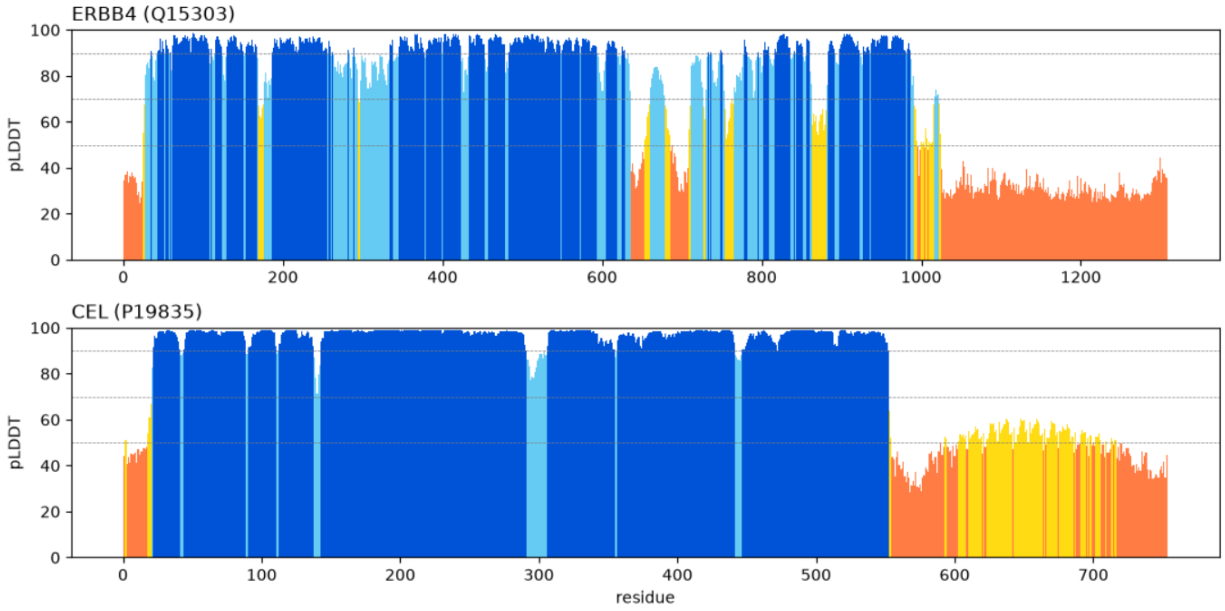
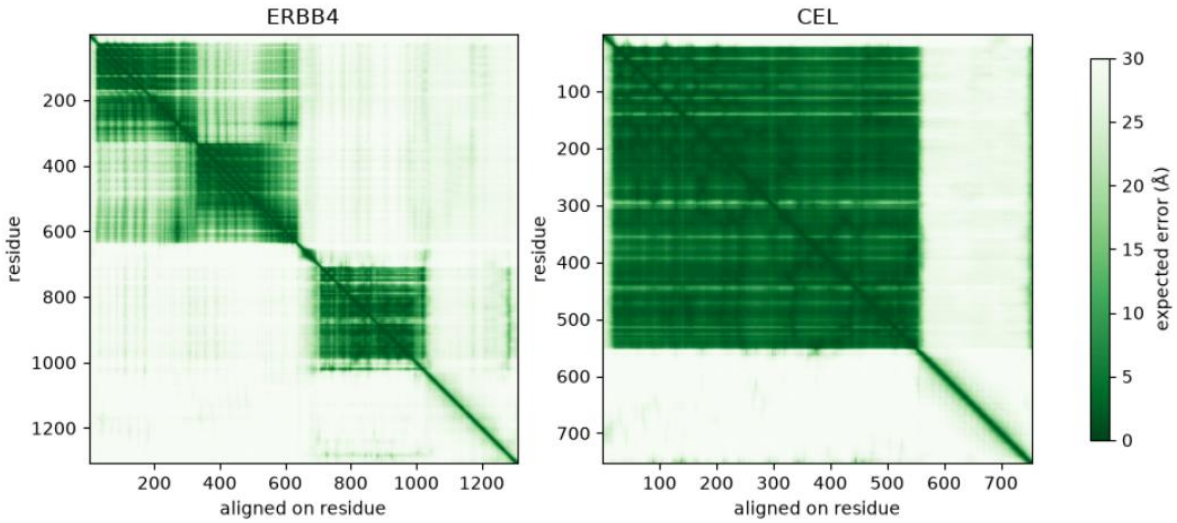


FIGURE: ERBB4 and CEL PAE heatmap




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ERBB4: 2 confident domain(s); mean PAE between domains 1 and 2: 28.7 Å
 start  end  confident_residues
    27  635                601
   711 1022                253

CEL: 1 confident domain(s); mean PAE between domains 1 and 2: nan Å
 start  end  confident_residues
    21  552                532
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CEL

The CEL AlphaFold model has a mean pLDDT of 82.6, suggesting better overall confidence than ERBB4. About 66.3% of the model has a pLDDT above 90, while 16.2% is below 50. The lower-confidence regions are concentrated in the C-terminal variable number tandem repeat (VNTR) region, which consists of repeated, proline-rich sequences that are heavily O-glycosylated. This tail is intrinsically disordered, so it does not have one stable structure that AlphaFold can confidently predict. The model also lacks the attached sugar groups, which are part of the native protein. The experimental structure 1F6W covers only residues 1–538 and excludes this tail, consistent with its lack of a stable, well-defined fold. Frameshift mutations that truncate the VNTR

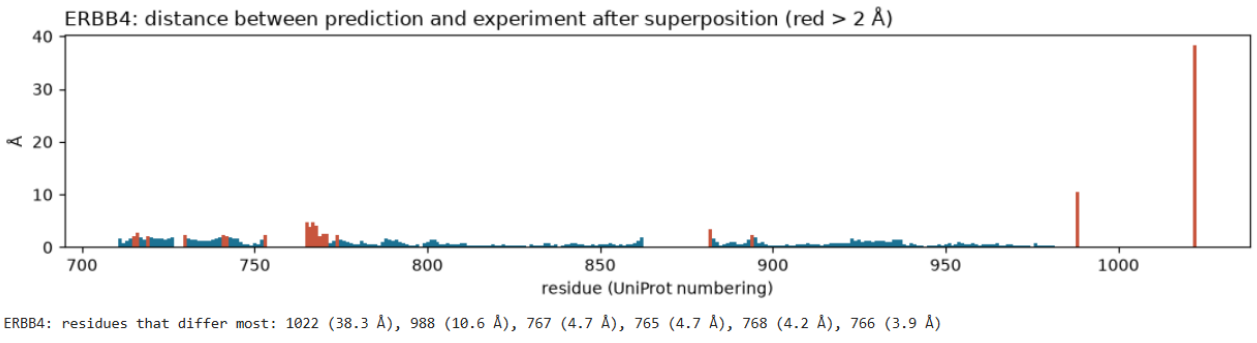
region have also been associated with MODY8, showing that this region is biologically important even though it is disordered. CEL has one confidently modelled domain, so its catalytic region is more suitable for structural assessment than its repetitive C-terminal tail.

2. How well does each prediction match experiment?

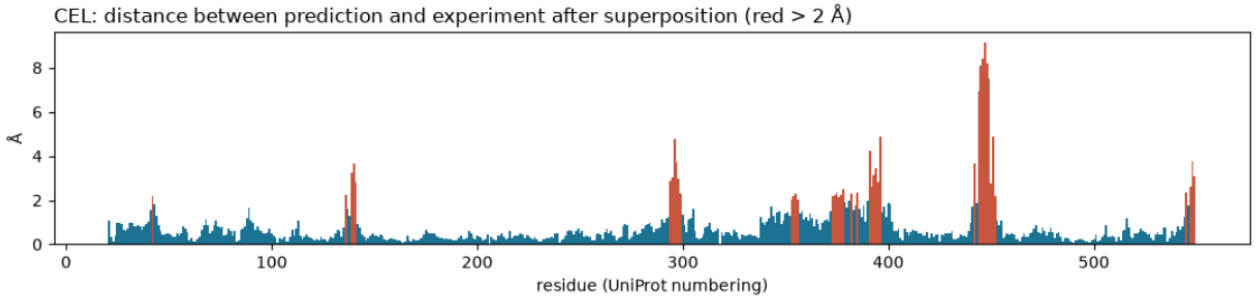
ERBB4 vs 2R4B (chain A)

The predicted structure agrees well with the experimental structure across most of the kinase domain. The largest deviations occur at residues 1022 (38.3 Å) and 988 (10.6 Å), while residues 765–768 show smaller deviations of approximately 4–5 Å. Residues 988 and 1022 lie beyond the kinase domain, toward the intrinsically disordered C-terminal tail identified in Question 1. This agreement between the confidence analysis and deviation plot suggests that the largest differences occur in a region where the predicted structure is already considered unreliable. Residues 765–768 are not among the 22 listed inhibitor-contact residues, but they lie within the broader inhibitor-site region, so their deviations could reflect local conformational differences. The experimental structure 2R4B contains a bound inhibitor, whereas the AlphaFold model does not, meaning ligand-induced changes in pocket shape may also contribute to the differences. Overall, the kinase core remains reasonably well supported, although the exact pocket conformation should be interpreted cautiously.

[FIGURE: ERBB4 and CEL per-residue deviation after superposition



ERBB4: residues that differ most: 1022 (38.3 Å), 988 (10.6 Å), 767 (4.7 Å), 765 (4.7 Å), 768 (4.2 Å), 766 (3.9 Å)



CEL: residues that differ most: 447 (9.1 Å), 446 (8.4 Å), 448 (8.2 Å), 445 (8.1 Å), 449 (7.5 Å), 444 (6.9 Å)

CEL vs 1F6W (chain A)

The largest deviations occur at residues 444–449, reaching approximately 7–9 Å, close to the catalytic residue His455. The experimental structure 1F6W has no bound ligand, so this difference may reflect the flexibility of a surface loop near the active site. A substrate-free crystal structure captures one conformation, while AlphaFold may predict a different plausible conformation. Smaller deviations near residues 140 and 295 also approximately overlap with mobile surface loops described in the experimental study. These differences do not necessarily indicate prediction errors; they may reflect genuine flexibility or differences between the crystallized and predicted conformations. The overall RMSD of 1.35 Å indicates good structural agreement, but the localized deviations near the catalytic region warrant caution when interpreting CEL's potential for inhibitor binding.

Note: 1F6W carries two surface mutations (N186D and A298D) introduced to aid crystallization, which appear as mismatches in the sequence alignment.

3. Pocket assessment and druggability calls

Pocket evidence

Target	Pocket	Rank	Druggability score	mean pLDDT	PAE within (Å)	Verdict	Note
ERBB4	93	1	0.894	80.9	5.4	keep	Covers 68% of GW7 inhibitor site
CEL	56	1	0.925	93.7	2.1	keep	Not the catalytic site
CEL	—	4	0.135	95.6	2.2	keep	Contains Ser214 and His455
CEL	—	13	0.024	94.5	1.8	keep	Contains Ser214

Confidence rules applied: reject if mean pLDDT < 70; suspect if PAE among lining residues > 10 Å; otherwise keep.

ERBB4 — Structure supports a small-molecule pocket

I kept ERBB4 pocket 93 because it passed the structural confidence checks and showed good agreement with the known inhibitor-binding site. The pocket ranked first, with a score of 0.894, a mean pLDDT of 80.9 and a within-pocket PAE of 5.4 Å. These values support the local reliability of the pocket. It covered 68% of the known GW7 inhibitor-site residues, which provides additional evidence that the predicted pocket corresponds to a real binding site. Although the inter-domain PAE was higher at 28.7 Å, this reflects uncertainty in the relative positioning of the larger domains rather than necessarily affecting the local pocket structure. Therefore, I would keep ERBB4 as a promising target for further investigation, while recognizing that the pocket does not perfectly reproduce the experimental binding site.

CEL — Possible but uncertain

CEL was less straightforward to assess. Its highest-scoring pocket had a score of 0.925 and high structural confidence, but it did not correspond to the catalytic site. The catalytic pocket was identified at rank 4 and included Ser214 and His455, two residues associated with the catalytic triad, but its druggability score was only 0.135. This shows that a high pocket score does not automatically mean that the pocket is functionally relevant, and a low score does not necessarily mean that a site cannot bind a drug. The experimental structure 1F6W also had no bound small-molecule ligand for direct validation. I would therefore classify CEL as possible but uncertain and keep it as a secondary target until further experiments establish whether its catalytic site can be targeted effectively.

4. Comparison with the Module 2 ranking and Module 3 priors

In Module 2, ERBB4 ranked 4th, while CEL ranked 11th. In Module 3, CEL had a slightly higher small-molecule tractability score of 0.72 compared with 0.70 for ERBB4, so the sequence-based analysis slightly favored CEL. However, after looking at the structures, I would prioritize ERBB4 because its predicted pocket overlaps with the known inhibitor-binding site, covering 68% of the known contact residues. In comparison, CEL's highest-scoring pocket is not its catalytic site, and the catalytic pocket has a low druggability score of 0.135. This shows me that sequence-based predictions alone are not enough to determine whether a target has a useful drug-binding site. I would keep both targets for further investigation, but focus on ERBB4 first because its structural evidence is stronger. I would keep CEL as a secondary target until more experiments can confirm whether its catalytic site can be targeted effectively.

5. Limitations that apply to these targets

Limit 1: one structural snapshot — ERBB4

One limitation of my ERBB4 analysis is that AlphaFold predicts only one structure, but the kinase can adopt different conformations depending on its activity and whether an inhibitor is bound. My analysis used the inhibitor-bound structure 2R4B for comparison, so the predicted pocket might not represent the binding site in every conformation. To investigate this, I would compare my model with experimental ERBB4 structures in different conformations, including 3BBT, which contains lapatinib in the inactive form. Comparing the pocket shapes would help me understand how the binding site changes and which conformation might be more suitable for inhibitor binding.

Limit 2: pocket-scoring limitations — CEL

For CEL, the catalytic pocket is confidently modelled, but its druggability score is only 0.135. I think the scoring method may underestimate this site because it is relatively open and polar rather than a deep, enclosed pocket. However, the inhibitor-bound structures 6H18 and 6H0V show that compounds can bind covalently to the catalytic serine, suggesting that the low score does not necessarily mean the site cannot be targeted. As a next step, I would compare these experimental structures with my CEL model to examine how the inhibitors interact with the catalytic site. I would then test promising compounds using an enzyme activity assay to determine whether they actually inhibit CEL.

6. Ligand-site assessment

For ERBB4, I compared the predicted pocket with the GW7 inhibitor-binding site in experimental structure 2R4B, where the inhibitor is bound to chain A. Pocket 93 covered 68% of the 22 inhibitor-contact residues. This supports the idea that the predicted pocket overlaps with the known binding site, but the incomplete coverage indicates that the two structures do not match perfectly. One possible explanation is induced fit: inhibitor binding can change the shape of the pocket compared with the unbound predicted structure. In addition, Cys803 lies near the solvent-exposed edge of the binding site, where the pocket may be more open and less likely to be identified as part of a single enclosed pocket by the pocket-detection algorithm. I would therefore consider the ERBB4 site a reasonable candidate for inhibitor binding, while treating the incomplete coverage as a limitation.

For CEL, the experimental structure 1F6W did not contain a bound small-molecule ligand, so I could not perform the same direct comparison between a predicted pocket and an experimentally observed ligand-binding site. Although the catalytic site contains relevant catalytic residues, its low druggability score and the lack of a ligand in 1F6W mean that its ability to bind a small-molecule inhibitor cannot be established from this analysis alone. Experimental inhibitor-bound structures and enzyme activity assays would provide stronger evidence for CEL's druggability.

GW7 (chain A) contact site in UniProt numbering: 724, 732, 749, 750, 751, 772, 781, 782, 783, 794, 795, 796, 797, 798, 799, 802, 803, 850, 860, 861, 862, 864.

Anchor check: p53

The RMSD between the p53 predicted structure and the experimental structure was 0.68 Å across 192 confident residues (pLDDT ≥ 70) and 0.73 Å across all 194 residues, which shows that the structures agree very closely. However, the experimental structure contains three zinc ions and DNA, while the AlphaFold model has neither. From my hotspot table, C176, H179, C238 and C242 coordinate zinc at distances of 2.1–2.3 Å, so mutations at these positions could affect zinc binding and destabilize the protein's structure. R248 and R273 directly contact DNA at 3.4 Å and 2.5 Å, respectively, while R249 is close to DNA at 7.2 Å. These residues are more directly involved in DNA binding. Other hotspots, including R175, Y220, G245 and R282, have structural roles; for

example, R175 is 5.3 Å from zinc, close to the binding site but not directly coordinating it. This helps me understand the Module 3 variant scores better because a sequence-based score can predict that a mutation is damaging, but it does not explain exactly how it affects the protein. Structural analysis helps distinguish mutations that disrupt zinc binding and protein stability from those that interfere with DNA binding even when the protein remains folded.

Addendum: Disease and target choice; discordance and limitations

BIOT 6900 · Module 4 · Mohammed Sharukh. Added October 29, 2026, per the instructor's announcement asking every report to include these two sections. The analysis and numbers above are unchanged.

A. Disease, data and target choice, with biological interpretation

Disease and data. The disease is pancreatic ductal adenocarcinoma (PDAC), carried from Module 2 (CPTAC-PDAC, open access via LinkedOmics) through the Module 3 tractability priors. The structural data were the AlphaFold Database models AF-Q15303-F1 (ERBB4) and AF-P19835-F1 (CEL), with their PAE files, and two RCSB PDB structures. I chose the PDB entries for coverage, resolution and ligand content: 2R4B (ERBB4 kinase, 2.40 Å, bound to a covalent thienopyrimidine inhibitor) and 1F6W (CEL catalytic domain, 2.30 Å, no ligand). UniProt provided the sequences and domain annotation. All sources are open access.

Why these two targets. ERBB4 (Module 2 rank 4) and CEL (rank 11) had the two strongest small-molecule priors in Module 3 (p_sm 0.70 and 0.72), so together they test whether a sequence-based prior predicts a real structural pocket.

What the results mean biologically.

- **ERBB4 recovers known biology.** The top-ranked kept pocket (score 0.894, lining pLDDT 80.9) covers 68% of the inhibitor-contact residues in 2R4B, including Cys803, the residue covalent pan-HER inhibitors attack. This is the ATP site where clinical ErbB inhibitors bind. ERBB4's relative EGFR is an established drug target in pancreatic cancer (erlotinib with gemcitabine), so a druggable ErbB kinase pocket is a credible PDAC result.
- **CEL is the surprising hit.** Its catalytic domain is modelled very confidently (RMSD 1.35 Å to 1F6W), yet its best pocket (0.925) is *not* the catalytic site, while the pocket holding Ser214 and His455 scores only 0.135. Open, polar catalytic clefts often score poorly in fpocket. Biologically, CEL is an acinar digestive enzyme that is strongly lost in PDAC (Module 2 RNA −6.09, protein −2.71). It is a useful pipeline control rather than a therapeutic target.

B. Discordance and limitations

Discordant gene: ERBB4. In Module 2, ERBB4 mRNA falls in tumours (−2.57) while its protein rises (+1.33). My structural call assumes the full-length receptor kinase is present in tumour cells. If the protein increase reflects a cleaved intracellular fragment, a stabilized inactive pool, or non-tumour cells in bulk tissue, a druggable pocket would not translate into a useful target. Structure answers "can a drug bind?", not "is the target active in this tumour?". Immunohistochemistry with antibodies to the extracellular and C-terminal regions would resolve this.

Other discordances. (1) Prior versus structure: CEL had the *highest* small-molecule prior but the weakest functional-site pocket, so sequence-based tractability did not predict a usable site.

(2) Model versus experiment: ERBB4's 2.82 Å RMSD is driven by a few terminal residues (1022 at 38.3 Å, 988 at 10.6 Å); the kinase core agrees closely. The CEL loop at 444–449 deviates by up to about 9 Å next to His455, plausibly genuine flexibility in a ligand-free crystal.

What I can claim. ERBB4's kinase domain contains a confidently modelled pocket that matches an experimentally observed inhibitor site. CEL's catalytic domain is confidently modelled, but its functional site is not clearly a druggable pocket.

What I cannot claim. I cannot claim binding affinity or selectivity against EGFR and HER2; pockets in other conformations (AlphaFold gives one static, ligand-free snapshot); the arrangement of ERBB4's extracellular and kinase domains (inter-domain PAE 28.7 Å, with no membrane modelled); or that either target is required by PDAC cells, which needs knockdown or knockout experiments.